

Lab 3: Architectural Pattern Selection & Justification

Nitin Somisetty | PES2UG24CS327 | Problem Statement 20

Project: Hospital Bed & ICU Allocation Dashboard

Domain: Healthcare & Telemedicine

Architecture Selection: *Layered Architecture*

Reason 1 – Clear Separation of Clinical Concerns

The system handles three distinct concerns: presenting real-time occupancy data to staff, computing triage severity and coordinating ICU/ambulance transfer workflows, and persisting patient and bed records. A layered structure (Presentation, Business, Data) keeps the touch/dashboard interface, the clinical prioritization logic, and the database access isolated from one another. This mirrors the requirements table directly — FR-001 through FR-004 are business-layer rules, while NFR-001 (WebSocket propagation) is a presentation-layer concern and NFR-002 (access control/audit logging) spans the data and business layers.

Reason 2 – Consistency for a Single-Facility Operational System

Unlike a distributed, independently-scaled system, this dashboard runs as one cohesive application within a single ER department, shared by a small set of authorized roles (Triage Nurse, Hospital Administrator) plus one external integration point (the inter-facility ambulance service). A layered architecture avoids the operational overhead of microservices (service discovery, distributed transactions) while still being easy for a small team to build, test, and maintain — appropriate for the scope defined in the problem statement.

Security Advantage

The layered structure enforces that all patient and bed-status writes pass through the Business Layer before reaching the Data Layer. This gives a single, centralized point to apply NFR-002 role-based access control and to log rejected access attempts, since no component in the Presentation Layer is allowed a direct connection to the Hospital Database Component. Unauthorized dashboard requests are therefore intercepted before they can touch persisted patient data.

Performance Benefit

Keeping bed-status change detection and WebSocket broadcast logic inside the Business Layer, close to the Data Layer, minimizes the number of hops between a bed-status update and its propagation to connected ER dashboard clients — supporting the sub-1-second requirement in NFR-001. A microservices approach would introduce network latency between separately deployed services for what is fundamentally one tightly-coupled, time-critical workflow.

Component Diagram

